Autonomous UGV for Safe Navigation in Unstructured Environments

Project Overview

This project focuses on the development of an autonomous Unmanned Ground Vehicle (UGV) capable of navigating safely and efficiently through unstructured environments.

The system combines classical computer vision algorithms with deep learning techniques to perceive, interpret, and interact with its surroundings while maintaining a strong emphasis on operational safety.

The project is currently under active development.

Motivation

Autonomous navigation in unstructured environments remains a challenging problem due to:

- Irregular terrain.
- Dynamic obstacles.
- Incomplete environmental information.
- Real-time computational constraints.

The objective of this project is to investigate and implement perception and navigation algorithms capable of enabling reliable autonomous mobility under these conditions.

Hardware Platform

Computing Unit

- NVIDIA Jetson AGX Orin

Selected to provide sufficient computational power for real-time execution of perception algorithms, allowing rapid development and experimentation before focusing on computational optimization.

Vision System

- ZED X Stereo Camera

Used for:

- Stereo depth estimation.
- Visual SLAM.

- Terrain perception.
- Obstacle detection.

Vehicle Platform

Electric motorcycle-based UGV platform adapted for autonomous operation.

Software Architecture

The navigation stack is divided into multiple perception layers.

Visual SLAM

Used for:

- Localization.
- Mapping.
- Environment reconstruction.

Terrain Classification

A classical computer vision approach based on RANSAC is employed to identify traversable surfaces.

Semantic Segmentation

Deep learning models are used to obtain semantic understanding of the environment.

Safety Redundancy Layer

A neural network-based terrain classifier serves as a secondary validation mechanism to increase reliability and safety.

Obstacle Avoidance

Obstacle avoidance is currently under development.

The proposed approach uses depth information generated by the stereo vision system to identify obstacles and generate safe navigation decisions.

Current Development Status

Implemented:

- Hardware integration.

- ZED X SDK integration.
- Visual SLAM.
- Terrain classification using RANSAC.
- Neural-network-based terrain validation.
- Real-time perception visualization.

Under Development:

- Obstacle avoidance.
- Navigation policy generation.
- Path planning improvements.

Future Work:

- Autonomous waypoint navigation.
- Dynamic obstacle handling.
- Multi-sensor fusion.
- Robust outdoor testing.

Experimental Results

The repository includes videos demonstrating:

- Initial vehicle platform.
- Hardware integration process.
- ZED SDK functionality.
- Terrain classification algorithms.
- Real-time perception outputs.
- Development and field testing activities.

Design Philosophy

The current phase prioritizes development speed and system functionality over computational optimization.

The Jetson AGX Orin was intentionally selected to remove hardware limitations during experimentation, allowing focus on algorithmic development and system integration.

Optimization for embedded deployment is planned for future stages.

Conclusion

This project demonstrates the development of a modular autonomous navigation platform that combines classical computer vision techniques and deep learning methods to achieve safe navigation in unstructured environments.

The system continues to evolve toward a fully autonomous UGV capable of reliable operation in real-world conditions.